

* التيار المتردد (AC): هو تيار يعكس اتجاهه بشكل دوري مع مرور الزمن.

* المعادلات اللحظية (Instantaneous Values):

(1) الجهد اللحظي (I. voltage): $e = E_m \sin \omega t$ حيث

- e = الجهد اللحظي.
- E_m = القيمة العظمى للجهد.
- ωt = سرعة الزاوية $\times$ الزمن.

(2) التيار اللحظي (I. current): $i = I_m \sin \omega t$ حيث

- i = التيار اللحظي.
- I_m = القيمة العظمى للتيار.

* القوة الدافعة الكهربائية (EMF):

$E = N \cdot A \cdot B \cdot \omega$ حيث أنها تعتمد على

- N = عدد اللفات.
- A = المساحة.
- B = كثافة الفيض.
- ω = سرعة الزاوية.

* سرعة الزاوية والتردد:

$\omega = 2\pi f$ ، سرعة الزاوية.

• التردد f يعتمد على عدد الأقطاب P والسرعة N على عدد الثواني.

$f = \frac{P}{2} \times \frac{N}{60}$ ← * عدد الثواني لكل دورة.

* القيمة الفعالة (R.M.S Values): هي قيمة التيار المستمر الذي تنتج نفس الحرارة

في نفس الزمن. $I_{r.m.s} = \frac{I_m}{\sqrt{2}} = 0.707 I_m$

* القيمة المتوسطة (Average Values): $I_{avg} = \frac{2}{\pi} I_m \approx 0.636 I_m$

* عامل الشكل (Form Factor - k): هو النسبة بين القيمة الفعالة R.M.S والقيمة المتوسطة.

$k = \frac{I_{rms}}{I_{avg}} = 1.11$

* عامل القمة (Peak Factor): هو النسبة بين القيمة العظمى والقيمة الفعالة.

Peak Factor = $\frac{I_{max}}{I_{rms}} = \sqrt{2} = 1.414$

Examples

* Example (1), calculate: i] The frequency of the emf.

ii] Maximum value of emf.

iii] The value of emf when the coil plane makes an angle 45° with the field direction.

Sol

$A = 30\text{ cm} \times 20\text{ cm} \Rightarrow 600 \times 10^{-4} \text{ m}^2$, $N = 40$, $B = 0.075$, $N = 1500 \text{ rpm}$

1]

~~$\omega = 2\pi f$~~
 $\therefore f = \frac{P}{2} \times \frac{N}{60}$
 $= \frac{2}{2} \times \frac{1500}{60}$
 $f = 25$

2] $\therefore E = N \cdot A \cdot B \cdot \omega$

$\omega = 2\pi f$

$E = 40 \times 600 \times 10^{-4} \times 0.075 \times (2\pi \times 25)$

$E = 28.27 \text{ V}$

3]

$\therefore e = E_m \sin \theta$

$e = 28.27 \sin 45^\circ$

$e = 19.98 \text{ V}$

* Example (2); Find the instantaneous value for $v = 5 \sin (2\pi \times 10^4 t)$ at $t = 25 \mu\text{s}$

Sol

$V = 5 \sin (2\pi \times 10^4 (25 \times 10^{-6}))$

$V = 5 \sin \left(\frac{1}{2} \pi \right)$

$V = 5 \sin \left(\frac{1}{2} \times 180 \right)$

$V = 5$

* إذا وجد π في موقع زاوية لأي نسبة مثلثية فيجب التحويل 180 مكان π

* Example (3), Write the mathematical expression for a 50 Hz sinusoidal voltage having peak value 80 V

Sol

$V = V_m \sin \omega t \Rightarrow V = V_m \sin 2\pi f t$

$\Rightarrow V = 80 \sin 2\pi \times 50 t \Rightarrow V = 80 \sin 100\pi t$

* Example (4), Calculate the maximum value of emf generated in a coil which is rotating at 50 r.p.s in a uniform magnetic field of 0.8 wb/m^2 .

The coil is wound on a square former having 5 cm in length and is wound with 300 turns.

Sol

$A = 25 \times 10^{-4} \text{ m}^2$, $N = 300$, $B = 0.8$, $f = 50 \text{ Hz}$

$\therefore E_m = N \cdot A \cdot B \cdot \omega \rightarrow 2\pi f$

$E_m = 300 \times 25 \times 10^{-4} \times 0.8 \times 2\pi \times 50$

$E_m = 188.5 \text{ V}$

* Example (5), An alternating current is given by $i = 141.4 \sin 314t$. Find: i] The maximum value. القيمة العظمى
 ii] The Frequency. التردد
 iii] time period. الزمن الدوري
 iv] The instantaneous value at $t = 3 \text{ ms}$.

1] $i = 141.4 \sin 314t$ $\therefore i = I_m \sin \omega t$ $I_m = 141.4$	2] <u>Sol</u> $\omega = 314$ $2\pi f = 314$ $f = \frac{314}{2\pi}$ $f = 50 \text{ Hz}$	3] <u>Sol</u> $T = \frac{1}{f}$ $T = \frac{1}{50}$ $T = 0.02 \text{ s}$	4] <u>Sol</u> $\therefore i = 141.4 \sin 314t$ $\therefore t = 3 \text{ ms} = 3 \times 10^{-3} \text{ s}$ $\therefore i = 141.4 \sin(314 \times 3 \times 10^{-3})$ $i = 141.35 \text{ A}$ ★
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* Example (6), The equation of an alternating current is given by $i = 42.42 \sin 628t$. Determine:-

- A] Its maximum value. B] Its Frequency. C] Its r.m.s value.
 D] Its average value. E] Its Form Factor. F] Its peak factor.

Solution
 $\therefore i = 42.42 \sin 628t$

A] $I_m = 42.42 \text{ A}$

$\omega = 628 \Rightarrow 2\pi f = 628$

B] $f = \frac{628}{2\pi} = 99.44 \text{ Hz}$

C] $I_{\text{r.m.s}} = \frac{I_m}{\sqrt{2}} = \frac{42.42}{\sqrt{2}} = 29.995 \text{ A}$

D] $I_{\text{avg}} = \frac{2}{\pi} I_m = \frac{2}{\pi} \times 42.42 = 27.005$

E] $k = \frac{I_{\text{r.m.s}}}{I_{\text{avg}}} = \frac{29.995}{27.005} = 1.11$

F] $\text{Peak Factor} = \frac{I_m}{I_{\text{rms}}} = \frac{42.42}{29.995} = 1.414$

* An alternating current of Frequency 60 Hz has maximum value of 120 A . 1] Write down the equation for the instantaneous value. 2] Reckoning time from the instant the current is zero and becoming positive, find instantaneous value after $1/360 \text{ s}$ second. 3] Time taken reach 96 A for the first time.

* Solution

$\therefore \omega = 2\pi f = 2\pi \times 60$

$\omega = 120\pi$

1] $i = 120 \sin(120\pi t)$

2] $t = 1/360$
 $i = 120 \sin(120\pi \times \frac{1}{360})$

$i = 103.9 \text{ A}$

3] $i = 96 \text{ A}$

$96 = 120 \sin(120\pi t)$

$\frac{96}{120} = \sin(120\pi t)$

$120\pi t = \sin^{-1}(\frac{96}{120})$

$120\pi t = 0.927 \text{ rad}$

$t = \frac{0.927}{120\pi} = 0.00246 \text{ s}$ ★